

Task 1: Creating and Renaming Files/Directories

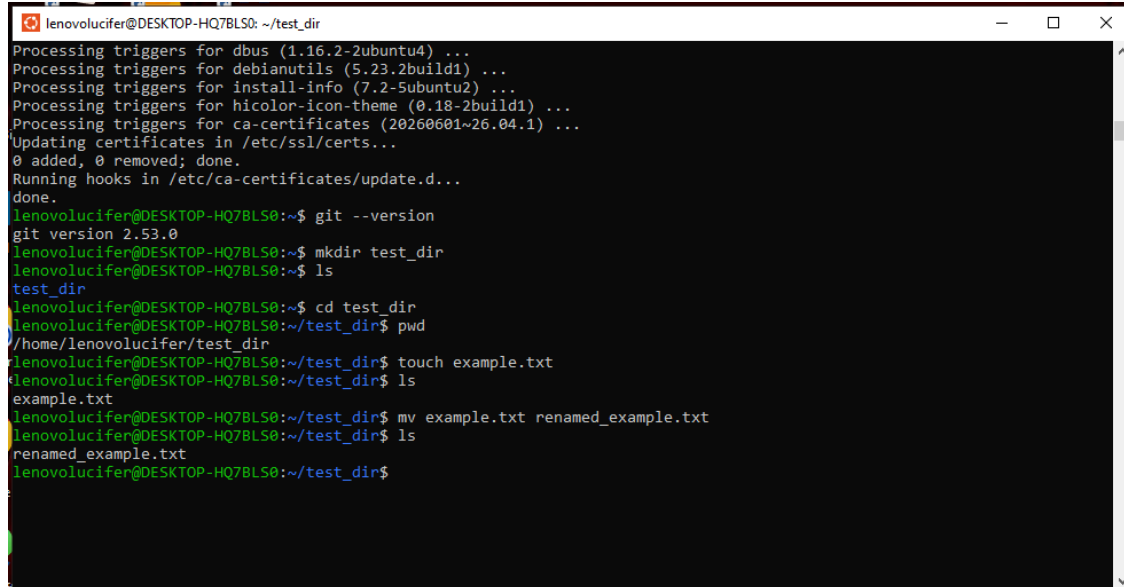
Commands Used

```
mkdir test_dir
cd test_dir
touch example.txt
mv example.txt renamed_example.txt
```

Explanation

- **mkdir** creates a new directory.
- **cd** changes the current directory.
- **touch** creates an empty file.
- **mv** renames or moves a file.

Screenshot

A terminal window titled 'lenovolucifer@DESKTOP-HQ7BLS0: ~/test_dir' showing the execution of several commands. The output includes system messages about processing triggers and updating certificates, followed by the user's commands and their results. The user checks the git version, creates 'test_dir', lists its contents (empty), changes into it, creates 'example.txt', and then renames it to 'renamed_example.txt'.

```
lenovolucifer@DESKTOP-HQ7BLS0: ~/test_dir
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for debianutils (5.23.2build1) ...
Processing triggers for install-info (7.2-5ubuntu2) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for ca-certificates (20260601~26.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
lenovolucifer@DESKTOP-HQ7BLS0:~$ git --version
git version 2.53.0
lenovolucifer@DESKTOP-HQ7BLS0:~$ mkdir test_dir
lenovolucifer@DESKTOP-HQ7BLS0:~$ ls
test_dir
lenovolucifer@DESKTOP-HQ7BLS0:~$ cd test_dir
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ pwd
/home/lenovolucifer/test_dir
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ touch example.txt
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ ls
example.txt
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ mv example.txt renamed_example.txt
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ ls
renamed_example.txt
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$
```

Task 2: Viewing File Contents

Commands Used

cat /etc/passwd

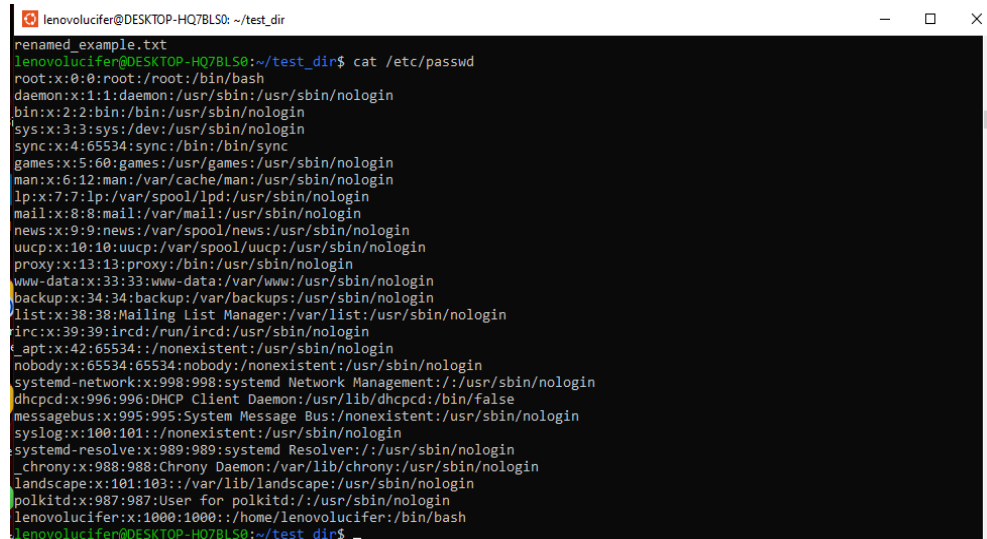
head -n 5 /etc/passwd

tail -n 5 /etc/passwd

Explanation

- **cat** displays the entire file content.
- **head** displays the first 5 lines.
- **tail** displays the last 5 lines.

Screenshot



```
lenovolucifer@DESKTOP-HQ7BLS0: ~/test_dir
renamed_example.txt
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
kapt:x:42:65534:/:nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101:/:nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:/:/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
lenovolucifer:x:1000:1000:~/home/lenovolucifer:/bin/bash
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$
```

```
lenovolucifer@DESKTOP-HQ7BLS0: ~/test_dir
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
lenovolucifer:x:1000:1000:/home/lenovolucifer:/bin/bash
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ tail -n 5 /etc/passwd
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
lenovolucifer:x:1000:1000:/home/lenovolucifer:/bin/bash
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ _
```

Task 3: Searching for Patterns

Command Used

grep "root" /etc/passwd

Explanation

grep searches for specific text patterns inside files. This command searches for the word "root" in /etc/passwd.

Screenshot

```
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
lenovolucifer@DESKTOP-HQ7BLS0:~/test_dir$ _
```

Task 4: Zipping and Unzipping

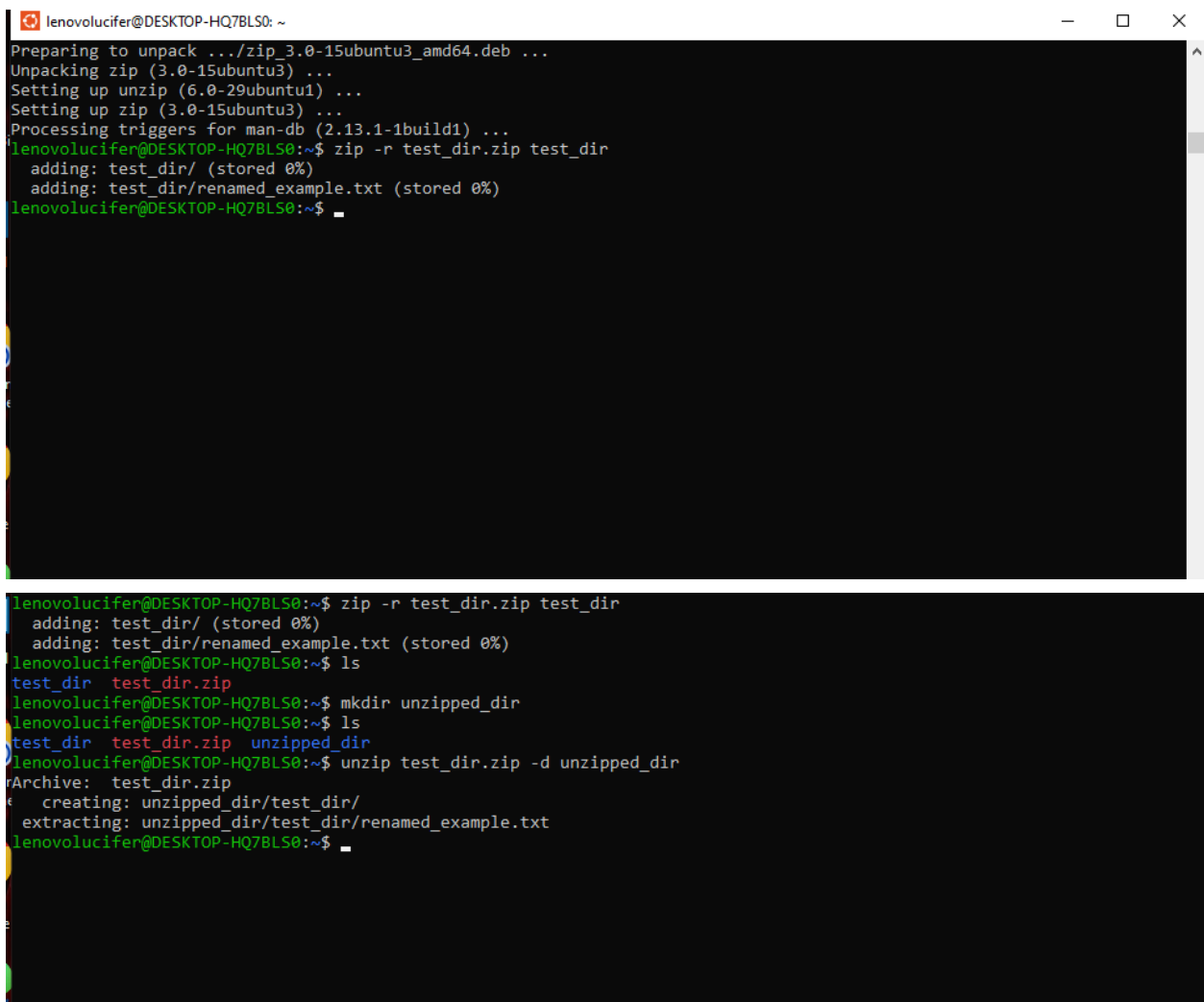
Commands Used

```
zip -r test_dir.zip test_dir
mkdir unzipped_dir
unzip test_dir.zip -d unzipped_dir
ls unzipped_dir/test_dir
```

Explanation

- **zip** compresses files and directories.
- **unzip** extracts compressed files.
- **ls** verifies extracted content.

Screenshot



```
lenovolucifer@DESKTOP-HQ7BLS0: ~
Preparing to unpack .../zip_3.0-15ubuntu3_amd64.deb ...
Unpacking zip (3.0-15ubuntu3) ...
Setting up unzip (6.0-29ubuntu1) ...
Setting up zip (3.0-15ubuntu3) ...
Processing triggers for man-db (2.13.1-1build1) ...
lenovolucifer@DESKTOP-HQ7BLS0:~$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
lenovolucifer@DESKTOP-HQ7BLS0:~$

lenovolucifer@DESKTOP-HQ7BLS0:~$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
lenovolucifer@DESKTOP-HQ7BLS0:~$ ls
test_dir  test_dir.zip
lenovolucifer@DESKTOP-HQ7BLS0:~$ mkdir unzipped_dir
lenovolucifer@DESKTOP-HQ7BLS0:~$ ls
test_dir  test_dir.zip  unzipped_dir
lenovolucifer@DESKTOP-HQ7BLS0:~$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
  creating: unzipped_dir/test_dir/
  extracting: unzipped_dir/test_dir/renamed_example.txt
lenovolucifer@DESKTOP-HQ7BLS0:~$
```

Task 5: Downloading Files

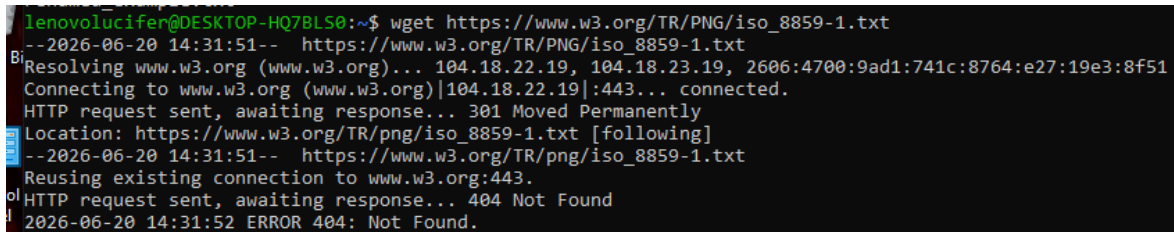
Command Used

wget <URL>

Explanation

wget is used to download files from the internet using a URL.

Screenshot



```
lenovolucifer@DESKTOP-HQ7BLS0:~$ wget https://www.w3.org/TR/PNG/iso_8859-1.txt
--2026-06-20 14:31:51-- https://www.w3.org/TR/PNG/iso_8859-1.txt
Resolving www.w3.org (www.w3.org)... 104.18.22.19, 104.18.23.19, 2606:4700:9ad1:741c:8764:e27:19e3:8f51
Connecting to www.w3.org (www.w3.org)|104.18.22.19|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.w3.org/TR/png/iso_8859-1.txt [following]
--2026-06-20 14:31:51-- https://www.w3.org/TR/png/iso_8859-1.txt
Reusing existing connection to www.w3.org:443.
HTTP request sent, awaiting response... 404 Not Found
2026-06-20 14:31:52 ERROR 404: Not Found.
```

Task 6: Changing Permissions

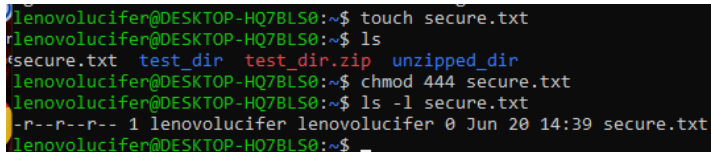
Commands Used

touch secure.txt
chmod 444 secure.txt
ls -l secure.txt

Explanation

- chmod changes file permissions.
- Permission 444 means read-only access for all users.

Screenshot



```
lenovolucifer@DESKTOP-HQ7BLS0:~$ touch secure.txt
lenovolucifer@DESKTOP-HQ7BLS0:~$ ls
secure.txt  test_dir  test_dir.zip  unzipped_dir
lenovolucifer@DESKTOP-HQ7BLS0:~$ chmod 444 secure.txt
lenovolucifer@DESKTOP-HQ7BLS0:~$ ls -l secure.txt
-r--r--r-- 1 lenovolucifer lenovolucifer 0 Jun 20 14:39 secure.txt
lenovolucifer@DESKTOP-HQ7BLS0:~$
```

Task 7: Working with Environment Variables

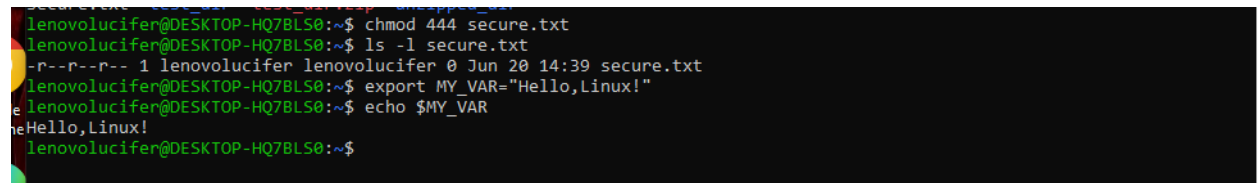
Commands Used

```
export MY_VAR="Hello, Linux!"  
echo $MY_VAR
```

Explanation

- `export` creates an environment variable.
- `echo` displays its value.

Screenshot



```
lenovolucifer@DESKTOP-HQ7BLS0:~$ chmod 444 secure.txt  
lenovolucifer@DESKTOP-HQ7BLS0:~$ ls -l secure.txt  
-r--r--r-- 1 lenovolucifer lenovolucifer 0 Jun 20 14:39 secure.txt  
lenovolucifer@DESKTOP-HQ7BLS0:~$ export MY_VAR="Hello, Linux!"  
lenovolucifer@DESKTOP-HQ7BLS0:~$ echo $MY_VAR  
Hello, Linux!  
lenovolucifer@DESKTOP-HQ7BLS0:~$
```